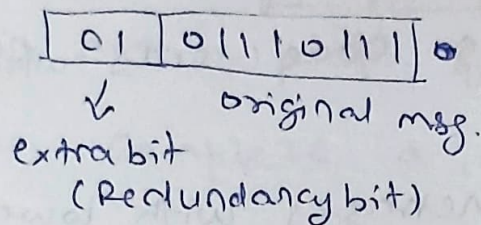


* Unit 2 *

* Source Coding Scheme for data compression *

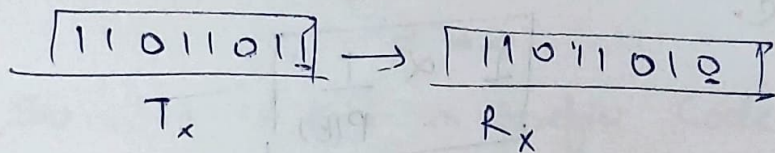
* Channel Coding Techniques -

1.) Redundancy / Extra bits (Parity bit) are added to the message bits, to be transmitted.



2.) Bandwidth requirement will increase.

3.) ^{used} For error detection & error correction



4.) Also known as Forward Error Control Coding.

5.) Examples are -

(a) Linear block codes

(b) Cyclic Code

(c) Hamming Code

(d) Convolution code

(e) Trellis Code

* Source Coding :-

Source Coding techniques,

- Redundancy / extra bits which are present in the message will reduce.
- It will reduce the bandwidth requirement.
- Variable length coding techniques such as Huffman to Shannon-Fano coding are used to give the code words with variable length.
- The symbols / messages with lower probability which contains more information, so more number of bits will require in comparison to the symbol with higher probability of occurrence.

$$I \propto \frac{1}{P(E)}$$

* Prefix :-

If the first K -bits of the codeword 'b' is exactly to the ~~the~~ codeword 'a' then 'a' is known as prefix of 'b'.

→ A code in which no code word is a prefix of another codeword is known as Prefix Code.

→ Prefix code is also known as

- 1) Prefix free Code
- 2) Instantaneous Code
- 3) Huffman Code.

Suppose a discrete memoryless (DMS) source is generating 3 symbols a, b & c

a $\rightarrow$ 0

b $\rightarrow$ 10

c $\rightarrow$ 110

'a' is not a prefix of any other codeword

'b' is not a prefix of any other codeword

'c' is not a prefix of any other codeword.

Then complete a, b & c is "prefix Code"

$C =$ a $\rightarrow$ 0

a $\times$

b $\rightarrow$ 1

b is prefix of c & d.

c $\rightarrow$ 10

c $\rightarrow$ x

d $\rightarrow$ 11

d $\rightarrow$ x

So it is not a prefix Code.

Techniques to determine whether a code is a prefix code or not.

1. Draw a binary tree which starts from a root node
2. Make the two branches such that left branch will represent binary zero & right branch will represent binary '1' ⁽⁰⁾
3. If all the symbols/letters are present at the external node then it is known as prefix code.
4. If any letter of the is present as internal node then it is not a prefix code.

Q Find out whether the given code is prefix code or not? $a \rightarrow '0'$ $b \rightarrow '1'$ $c \rightarrow '10'$ $d \rightarrow '11'$

Soln

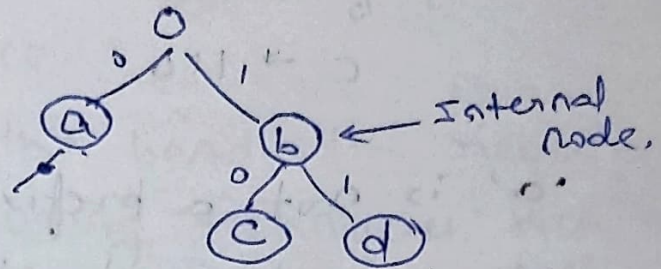
Sol

$a \rightarrow 0$

$b \rightarrow 1$

$c \rightarrow 10$

$d \rightarrow 11$



So It is not a prefix code.

Q2 Find out whether the given code is prefix code or not?

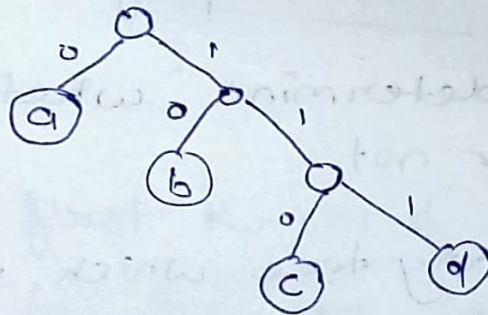
Soln

$a \rightarrow 0$

$b \rightarrow 10$

$c \rightarrow 110$

$d \rightarrow 111$



all are external nodes-

So given code is prefix code.

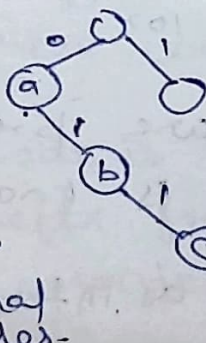
Q3 Prefix code or Not? $a \rightarrow 0$

Soln

$b \rightarrow 01$

$c \rightarrow 011$

$d \rightarrow 0111$



a b & c

are internal nodes.

It is not a prefix code.

Q4 A discrete memoryless source a, b, c

$$a \rightarrow 0$$

$$b \rightarrow 1$$

$$c \rightarrow 00$$

Then decode the codeword 100?

Solⁿ

$$\begin{array}{ccc} 1 & 0 & 0 \\ \hline b & c & \end{array}$$

or baa $\Rightarrow$ Not a prefix code

So it shows ambiguity.

So Transmitter side ensure the prefix code property before Transmission.

Note Fixed length code is always a prefix Code

Ex

$$a \rightarrow 101$$

$$b \rightarrow 001$$

$$c \rightarrow 010$$

$$d \rightarrow 111$$

} Fixed length code.

Q A discrete memory Source (DMS) generate 3 message having probabilities

$$P_1 = 0.7$$

$$P_2 = 0.2$$

$$P_3 = 0.1$$

Calculate 1) H 3) η (efficiency)

2) H_{max} 4) Redundancy (R)

Solⁿ

$$H = \sum P_i \log_2 \left(\frac{1}{P_i} \right)$$

$$H = 0.7 \log_2 \left(\frac{1}{0.7} \right) + 0.2 \log_2 \left(\frac{1}{0.2} \right) + 0.1 \log_2 \left(\frac{1}{0.1} \right)$$

$$H = 0.7 \frac{\log_{10}(0.7)}{\log_{10}(2)} + 0.2 \frac{\log_{10}(0.2)}{\log_{10}(2)} + 0.1 \frac{\log_{10}(0.1)}{\log_{10}(2)}$$

$$H = \frac{(0.7 \times 0.154 + 0.2 \times 0.698 + 0.1 \times 1)}{\log(2)}$$

$$H = \frac{0.1078 + 0.1396 + 0.1}{0.3010}$$

$$H = \frac{0.3474}{0.3010}$$

$$H = 1.154 \text{ bit/sym}$$

$$\begin{aligned} \text{(ii) } H_{\max} &= \log_2(M) \\ &= \log_2(3) \\ &= 1.58 \text{ bits/sym.} \end{aligned}$$

$$\text{(iii) } \eta = \frac{H}{H_{\max}} \times 100 = \frac{1.154}{1.58} \times 100$$

$$\eta = 0.73 \times 100$$

$$\boxed{\eta = 73\%}$$

$$\begin{aligned} \text{iv) Redundancy} &= 100 - \eta \quad (1 - \eta \text{ in points}) \\ &\Rightarrow 100 - 73 \end{aligned}$$

$$\Rightarrow 27\%$$

Ans

* Shannon Fano Coding

Steps to solve the problems.

- a) List the symbols in the decreasing order of probability.
- b) Divide the complete section into two sub sections such that both the sections are nearly equiprobable.
- c) ~~Assign divide both the sections according to the rule listed in the step-2~~
- d) Assign binary zero '0' to the upper section & assign binary '1' to the lower section.
- e) Again divide both the sections according to the rule listed in the step-2.
- f) Allocate or assign binary zero '0' to the upper half of each section & binary one to the lower half of each section.
- g) Repeat the steps until further division is not possible.

Q Find the entropy & efficiency of the source which generates four symbols $S_1 \rightarrow S_4$ with the probabilities $S_1 = 1/2, S_2 = 1/4, S_3 = 1/8, S_4 = 1/8$ by using Shannon fano coding.

Solⁿ Arrange in descending order

$$\frac{1}{4} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}$$

$$S_1 \rightarrow 1/2 \rightarrow 0$$

$$S_2 \rightarrow 1/4 \rightarrow 10$$

$$S_3 \rightarrow 1/8 \rightarrow 110$$

$$S_4 \rightarrow 1/8 \rightarrow 111$$

	codeword	length of code
S_1	0	1
S_2	10	2
S_3	110	3
S_4	1110	4

$$H(x) = \sum_{i=1}^4 P_i \log\left(\frac{1}{P_i}\right)$$

$$H(x) = \frac{1}{2} \log(2) + \frac{1}{4} \log(4) + \frac{1}{8} \log(8) + \frac{1}{8} \log(8)$$

$$H(x) = \frac{1}{2} \times 1 + \frac{1}{4} \times 2 + \frac{1}{8} \times 3 + \frac{1}{8} \times 3$$

$$H(x) = \frac{1}{2} + \frac{1}{2} + \frac{3}{8} + \frac{3}{8}$$

$$H(x) = \frac{7}{4} = 1.75 \text{ bit/symbol.}$$

$$\therefore \eta = \frac{H(x)}{\hat{H}} \quad \text{where } \hat{H} = \sum_{i=1}^n P_i n_i$$

$$\begin{aligned} \hat{H} &= \frac{1}{2} \times 1 + \frac{1}{4} \times 2 + \frac{1}{8} \times 3 + \frac{1}{8} \times 3 \\ &= 1.75 \text{ bits/symbol.} \end{aligned}$$

$$\eta = \frac{1.75}{1.75} \times 100 = 100\%$$

Q Find the codeword for the eight symbols $S_1 \rightarrow S_8$ with the probabilities

$\{ \frac{1}{4}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}, \frac{1}{16}, \frac{1}{16}, \frac{1}{16}, \frac{1}{16} \}$ also find out efficiency & redundancy by using shanon fano coding.

Soln

					Codeword	Length
S_1	$\frac{1}{4}$	0	0		00	2
S_2	$\frac{1}{4}$	0	1		01	2
S_3	$\frac{1}{8}$	1	0	0	100	3
S_4	$\frac{1}{8}$	1	0	1	101	3
S_5	$\frac{1}{16}$	1	1	0	0	4
S_6	$\frac{1}{16}$	1	1	0	1	4
S_7	$\frac{1}{16}$	1	1	1	0	4
S_8	$\frac{1}{16}$	1	1	1	1	4

$$S_1 + S_2 = \frac{1}{2}$$

$$S_3 + S_4 + S_5 + S_6 + S_7 + S_8 = \frac{1}{2}$$

$$H(x) = \sum_{i=1}^8 P_i \log\left(\frac{1}{P_i}\right)$$

$$= \frac{1}{4} \log(4) + \frac{1}{4} \log(4) + \left(\frac{1}{8} \log(8)\right) 2 + \left(\frac{1}{16} \log(16)\right) 4$$

$$= \frac{1}{4} \times 2 + \frac{1}{4} \times 2 + \frac{1}{8} \times 3 \times 2 + \frac{1}{16} \times 4 \times 4$$

$$= 2.75 \text{ bits/symbol}$$

$$\hat{H} = \sum_{i=1}^n P_i \log_2 \frac{1}{P_i}$$

$$= \frac{1}{4} \times 2 + \frac{1}{4} \times 2 + \frac{1}{8} \times 3 + \frac{1}{8} \times 3 + \frac{1}{16} \times 4 \times 4$$

$$= \frac{1}{2} + \frac{1}{2} + \frac{6}{8} + 1$$

$$= 2.75$$

$$\eta = \frac{H(X)}{\hat{H}} = \frac{2.75}{2.75} = 100\%$$

redundancy

$$r = 1 - \eta$$

$$r = 1 - 1 = 0$$

* Huffman Coding

Huffman Coding is a source coding technique that is used for data compression.

- It provides maximum efficiency η & minimum redundancy.
- It is also known as minimum redundancy code or optimum code.

Steps 1. arrange the given symbols in decreasing order of probability.

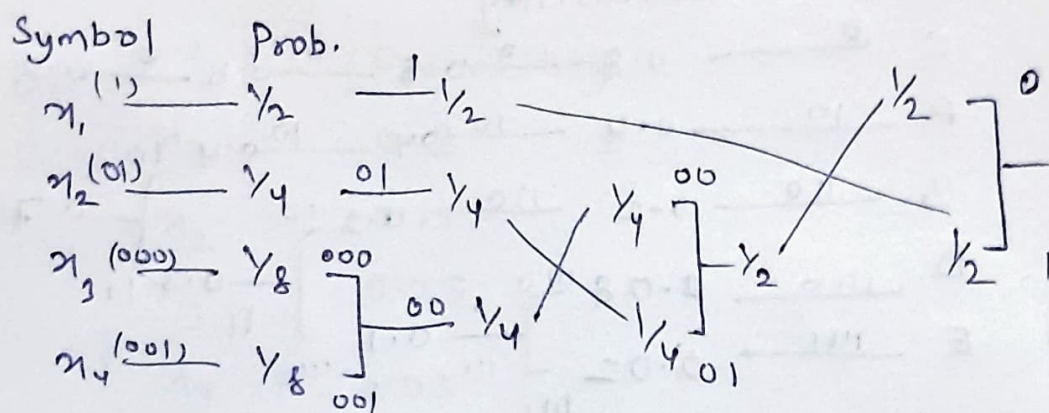
2. Combine the least probable ~~message~~ number of message & then assign them as high as possible.

3. Repeat the step 2 until there are two messages.

4. Rearrange the symbols in decreasing order at every step.

Q A source transmits different symbols x_1, x_2, x_3, x_4 with the probabilities given as $\{1/2, 1/4, 1/8, 1/8\}$. Find the code word of each symbol by using Huffman Coding & calculate efficiency of the system.

Solⁿ



So	Code word	no. of bits
x_1	1	1
x_2	01	2
x_3	000	3
x_4	001	3

$$H_m = P_i \log_2(1/P_i) = 1/2 \log(2) + 1/4 \log(4) + 2 \times 1/8 \log(8)$$

$$H_m = 1/2 + 1/2 + 6/8 = 14/8 = 1.75$$

$$H_m = 1.75 \text{ bit/symbol.}$$

$$\hat{H} = -\log_2 \sum p_i = 1/2 \times 1 + 1/4 \times 2 + 1/8 \times 3 \times 2$$

$$\hat{H} = 1.75 \text{ bit/symbol.}$$

$$\eta = \frac{H_m}{\hat{H}} = 100\%$$

Q. Decide Huffman code for the discrete codes of source independent symbol (A, B, C, D, E) with the probability given (0.4, 0.2, 0.8, 0.08, 0.02)

Solⁿ

Symbol	Probability
C 0	0.8
A 10	0.4
B 110	0.2
D 1110	0.08
E 1111	0.02

	bits
A 10	= 2
B 110	= 3
C 0	= 1
D 1110	= 4
E 1111	= 4

$$H_x = \sum P_i \log_2 (1/P_i)$$

$$H_x = 0.4 \log_2 (1/0.4) + 0.2 \log_2 (1/0.2) + 0.8 \log_2 (1/0.8) + 0.08 \log_2 (1/0.08) + 0.02 \log_2 (1/0.02)$$

$$H_x = 1.32 \times 0.4 + 2.32 \times 0.2 + 0.8 \times 0.32 + 0.08 \times 3.64 + 0.02 \times 5.64$$

$$H_x = 1.634$$

$$\hat{H} = 0.4 \times 2 + 0.2 \times 3 + 0.8 \times 1 + 0.08 \times 4 + 0.02 \times 4 = 2.6$$

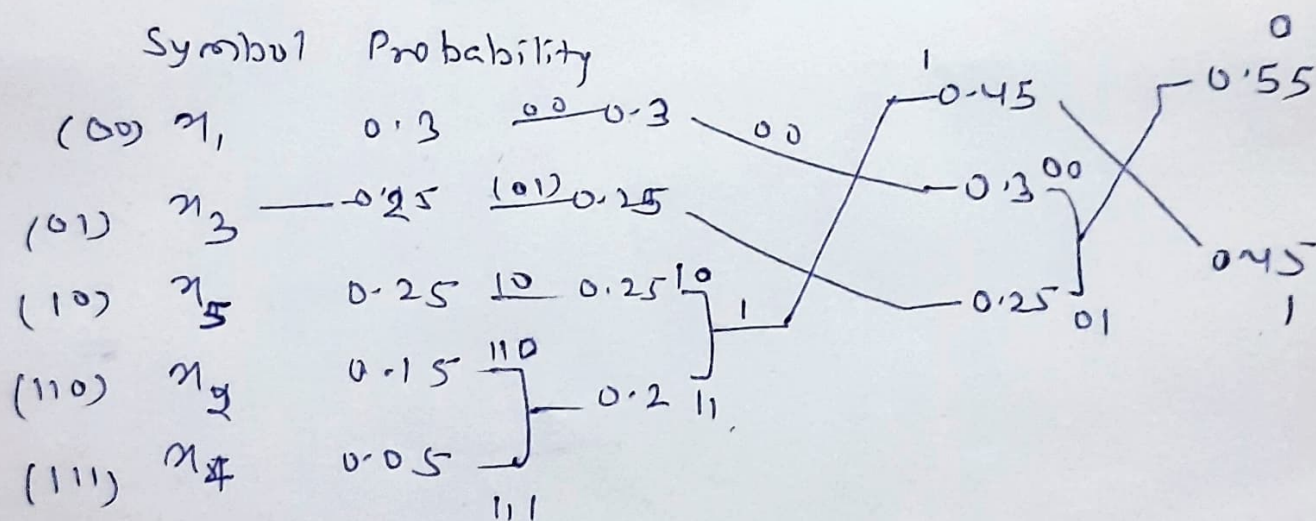
$$\eta = \frac{H_x}{\hat{H}} = \frac{1.634}{2.6} = 63\%$$

Q A Source transmit 5 different symbol x_1, x_2, x_3, x_4, x_5 with given probabilities (0.3, 0.15, 0.25, 0.05, 0.25) if the number of symbol used to code then symbol the 2. Then find the codeword using Huffman Coding.

Solⁿ

Symbol	Probability
x_1	0.3
x_2	0.15
x_3	0.25
x_4	0.05
x_5	0.25

Ans Descending order



$$H_x = 0.3 \times \log_2\left(\frac{1}{0.3}\right) + 0.15 \times \log_2\left(\frac{1}{0.15}\right) + 0.25 \times 2 \times \log_2\left(\frac{1}{0.25}\right) + 0.05 \log_2\left(\frac{1}{0.05}\right)$$

$$H_x = 2.135 \text{ bits}$$

$$\hat{H} = 0.3 \times 2 + 0.15 \times 3 + 0.25 \times 2 + 0.05 \times 3 + 0.25 \times 2$$

$$\hat{H} = 2.2$$

$$\eta = \frac{H\eta}{\hat{H}} = \frac{2.135}{2.2} \approx 97\%$$

$$\boxed{\eta = 97\%}$$